

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Module - I

Practical – IV

Roll No.:T054	Name:Priti Yadav
Class:TYIT	Batch:2
Date of Assignment: 20/08/26	Date/Time of Submission: 20/08/26

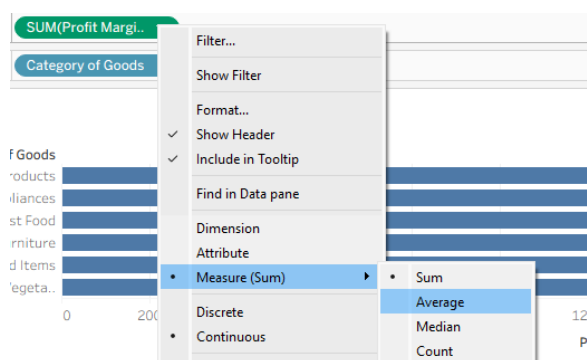
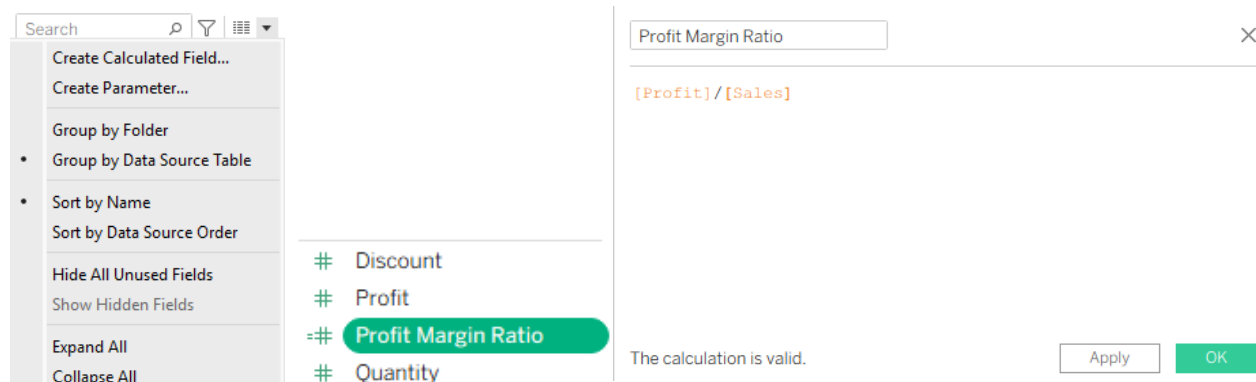
4: Performing Data Analysis using Calculations in Tableau

a) Create basic calculated fields using formulas.

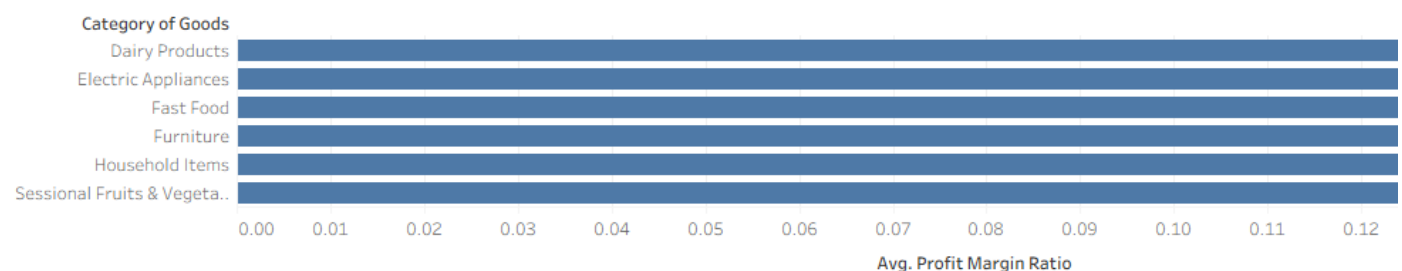
STEPS:-

1. Connect → store_sales_data (2).csv.
2. Create a Calculated Field
3. Enter the formula: $[\text{Profit}] / [\text{Sales}]$
4. A new Profit Margin Ratio field will appear in the Data pane.
5. Drag Category of Goods → Rows.
6. Drag Profit Margin → Columns.
7. It will create a chart showing the profit margin for each category.

OUTPUT:-



Profit Margin Ratio T054



b) Perform row-level calculations.

STEPS:-

1. Create a calculated field
2. Enter the name: Total Revenue Per Discount

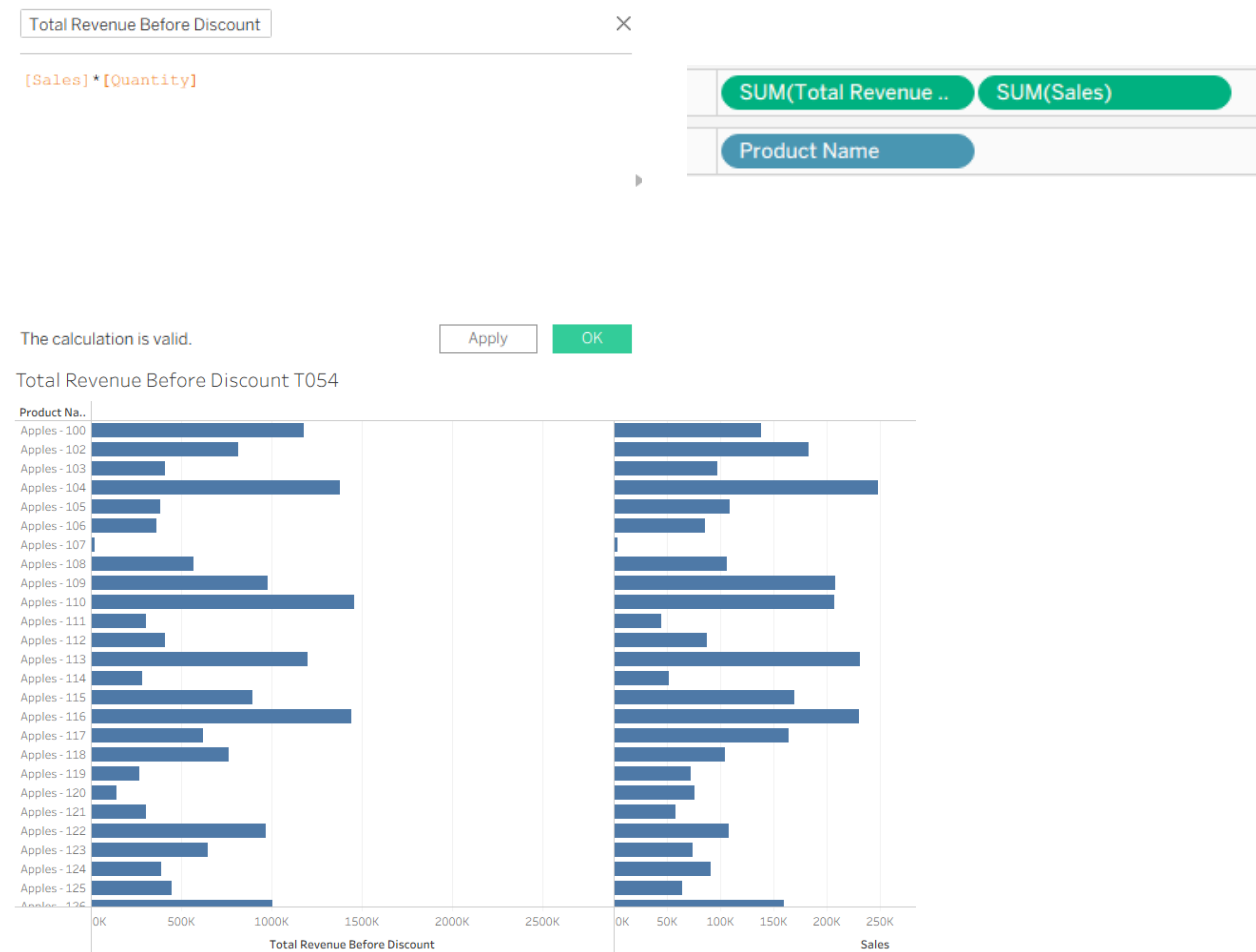
3. Enter the formula: $[\text{Sales}] / [\text{Quantity}]$

4. Drag Product Name → Rows.

5. Total Revenue Per Discount and Sales → Columns

6. It will create a chart showing the Total Revenue Per Discount for each category.

OUTPUT:-



c) Perform aggregate-level calculations.

STEPS:-

1. Create a calculated field

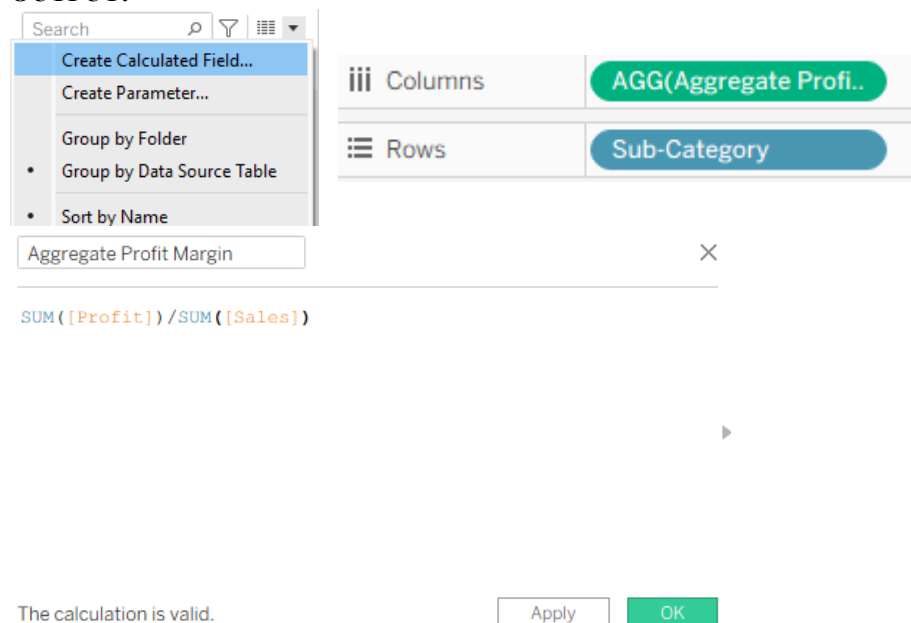
2. Name the field: Aggregate Profit Margin.

3. Enter the formula: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$

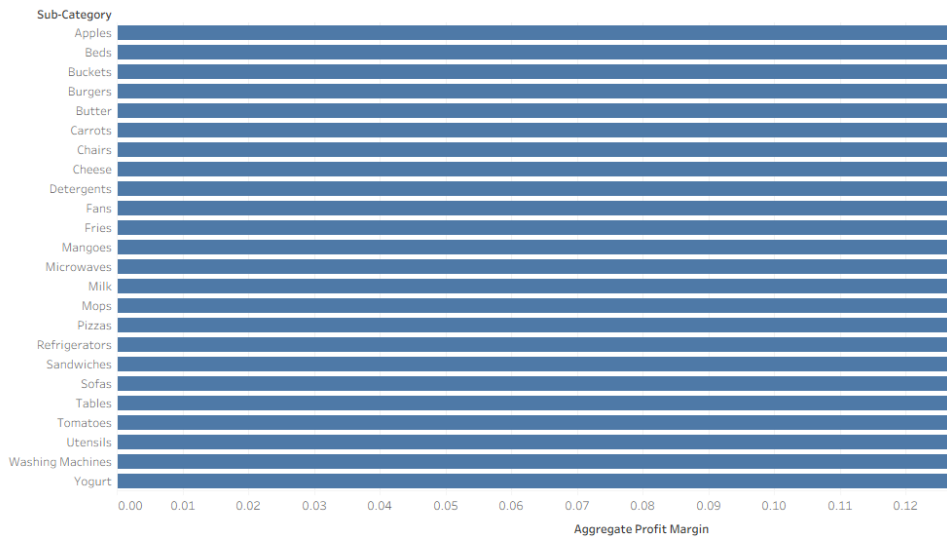
4. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf. Drag your newly created.

5. Aggregate Profit Margin calculated field and drop it onto the Columns shelf.

OUTPUT:-



Aggregate Profit Margin T054



d) Implement FIXED Level of Detail (LOD) calculations.

STEPS:-

1. Create a calculated field
2. Name the field: Fixed Regional Sales.
3. Enter the formula: `{ FIXED [State] : SUM([Sales]) }`
4. Drag the State dimension from the Data Pane and drop it onto the Rows shelf. 2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State. 3. Drag the raw Sales measure pill onto the Columns shelf.
5. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.

OUTPUT:-

Create Calculated Field...

Create Parameter...

Group by Folder

Group by Data Source Table

Sort by Name

Columns

SUM(Sales)

SUM(Fixed Regional ..)

Rows

State

Sub-Category

×

{ FIXED [State] : SUM ([Sales]) }

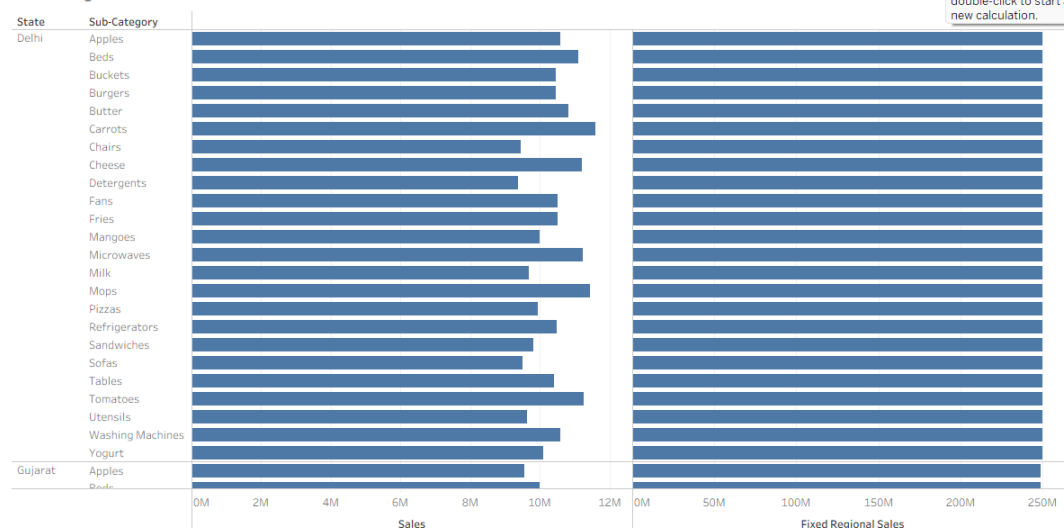
▶

The calculation is valid.

Apply

OK

Fixed Regional Sales T054



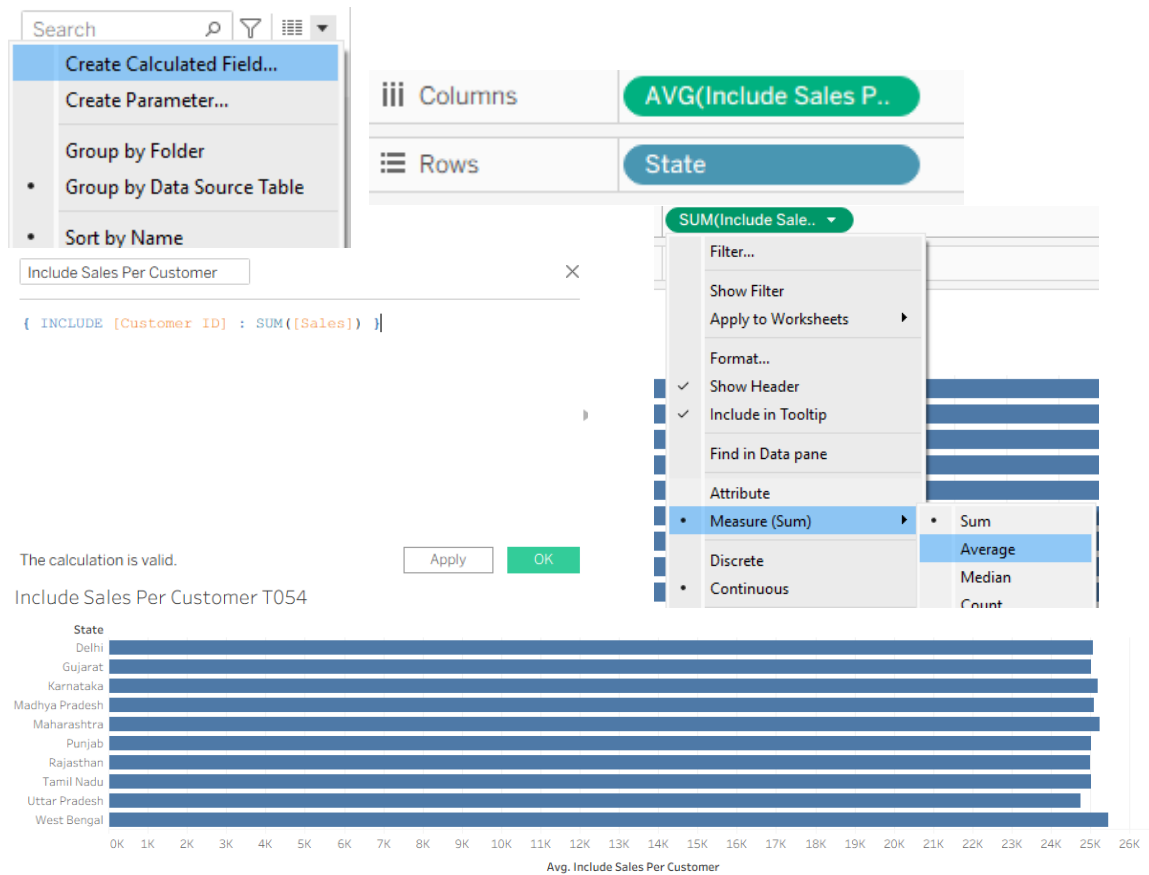
e) Use INCLUDE and EXCLUDE LOD expressions

STEPS:-

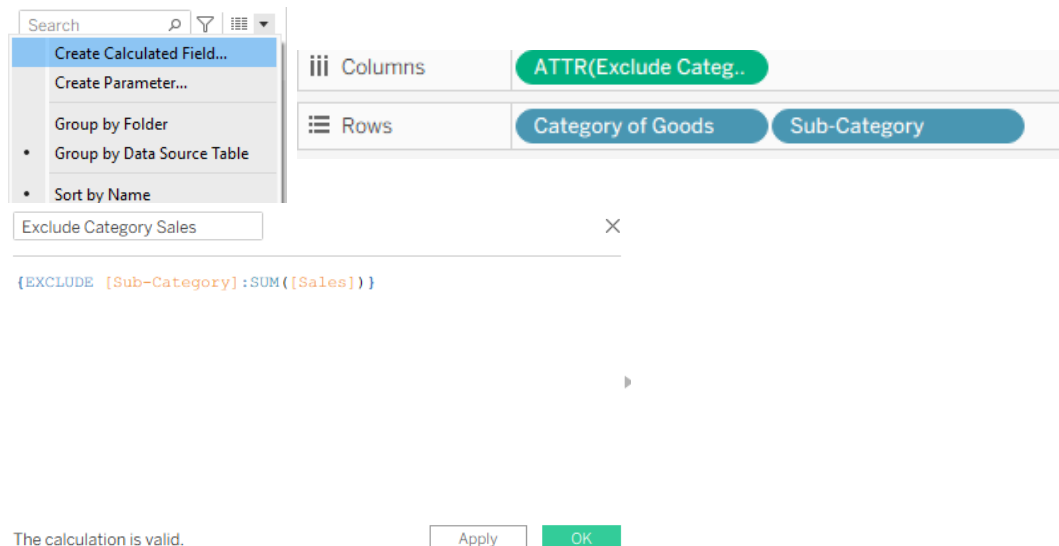
1. Create a calculated field
2. Name the field: Include Sales per Customer.
3. Enter the formula: { INCLUDE [Customer ID] : SUM([Sales]) }
4. Drag the State dimension from the Data Pane and drop it onto the Rows shelf. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.
5. Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.
6. Creating an EXCLUDE LOD Calculated Field.
7. Drag Category of Goods onto the Rows shelf.
8. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).
9. Create Calculated Field.
10. Name the field: Exclude Category Sales.
11. Enter the formula: { EXCLUDE [Sub-Category] : SUM([Sales]) }
12. Click OK and drag this new field onto the Columns shelf.

OUTPUT:-

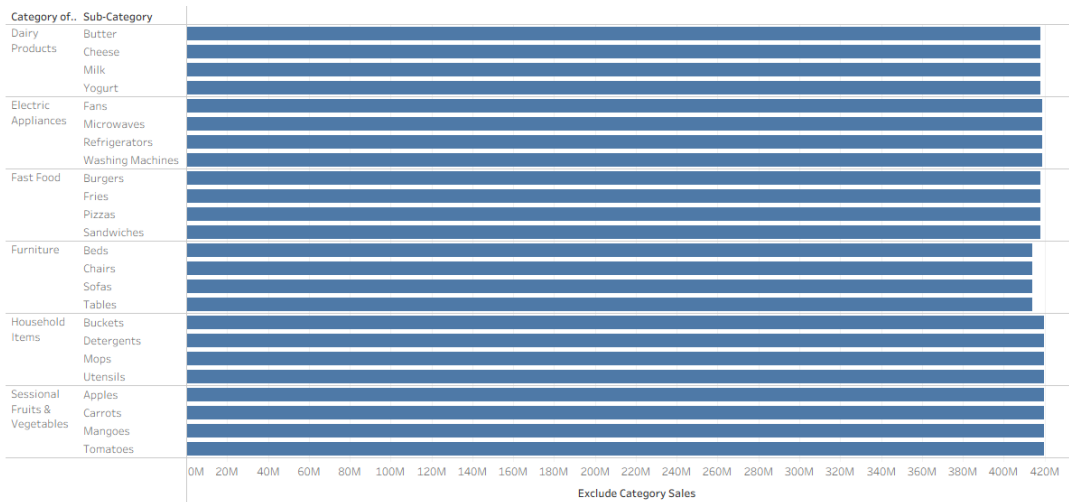
INCLUDE:



EXCLUDE:



Exclude Category Sales T054



f) Create and use parameters for dynamic analysis.

STEPS:-

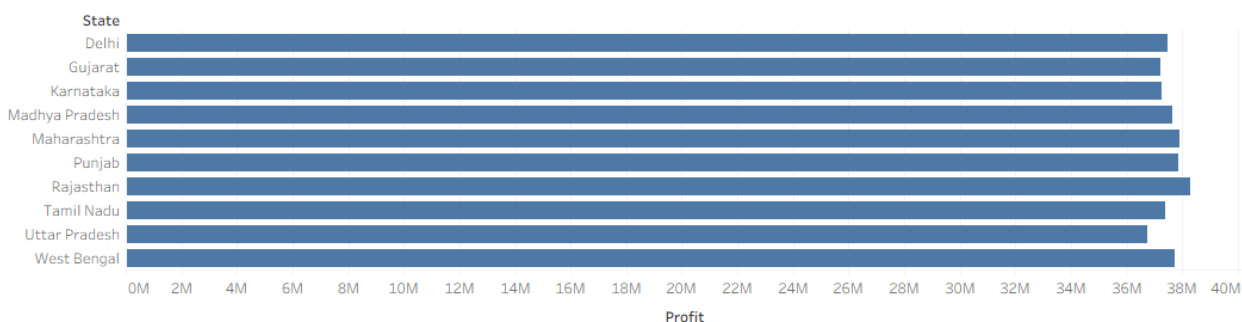
1. Create Calculated Field...
2. Name the field: Select Profit Threshold.
3. Set the Data Type to Float (or Integer).
4. Set the Current Value to 500.
5. Create Calculated Field...
6. Name this field: KPI Threshold Met?.
7. Input the following logical IF-THEN conditional statement:
IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved" ELSE "Below Target"
END
8. A text box slider will slide into the upper right corner of your sheet canvas view. Type a low value like 100 or a high value like 5000 into that parameter box and hit Enter.

OUTPUT:-

The screenshot shows the Power BI interface with the following elements:

- Search Bar:** Contains the text "Search".
- Columns:** Contains the calculated field "SUM(Profit)".
- Rows:** Contains the field "State".
- Create Parameter Dialog:**
 - Name:** Select Profit Threshold
 - Properties:**
 - Data type:** Float
 - Display format:** 500
 - Current value:** 500
 - Value when workbook opens:** Current value
 - Allowable values:** All (selected), List, Range
- KPI Threshold Met? Dialog:**
 - Name:** KPI Threshold Met?
 - Formula:** IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved" ELSE "Below Target" END
 - Message:** The calculation is valid.
 - Buttons:** Apply, OK

KPI Threshold Met T054



Sub-Category

Year

Measure Names

Aggregate Profit Margin

Discount

Exclude Category Sales

Fixed Regional Sales

Include Sales Per Cust...

KPI Threshold Met?

Profit

Profit Margin Ratio

Quantity

Sales

Total Revenue Before ...

Latitude (generated)

Parameters

Select Profit Threshold

Show Parameter

Cut

Copy

Edit...

Duplicate

Rename

Hide

Delete

Create

Default Properties

Folders

Replace References...

Describe...

AGG(KPI Threshold Me...

Target Achieved

Select Profit Thresh...

37000000

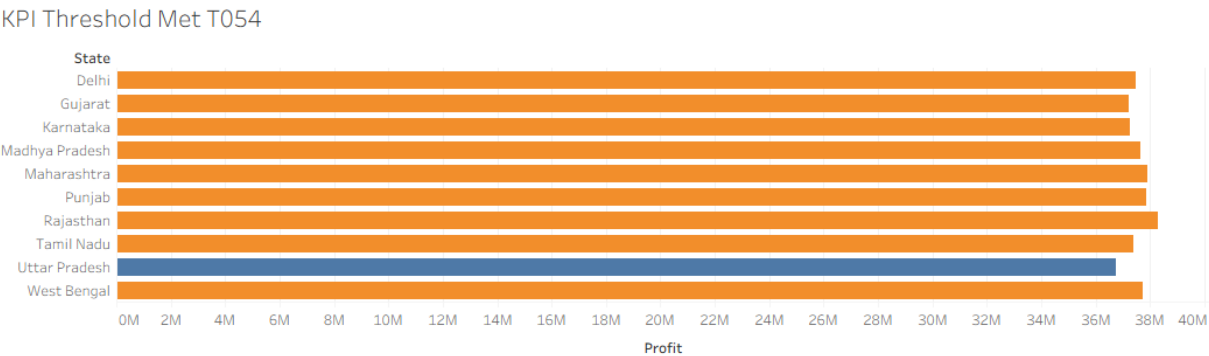
AGG(KPI Threshold Me...

Below Target

Target Achieved

Select Profit Threshold

3,70,00,000



- g) Apply calculations to fix data issues and enhance insights.**
- STEPS:**
- 1.Create Calculated Field.
 - 2.Name the field: Cleaned Discount.
 - 3.Input the following expression: `IFNULL([Discount], 0)`
 - 4.Create a new Calculated Field named Standardized City Type.
 5. Input the following text transformation expression: `UPPER([City Type])`
 6. Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.
 7. Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by right-clicking the pill.

Create Calculated Field...

Create Parameter...

Group by Folder

Group by Data Source Table

Sort by Name

Cleaned Discount

`IFNULL([Discount], 0)`

Create Calculated Field...

Create Parameter...

Group by Folder

Group by Data Source Table

Sort by Name

Standardized City Type

`UPPER([City Type])`

The calculation is valid.

Apply

OK

The calculation is valid.

Apply

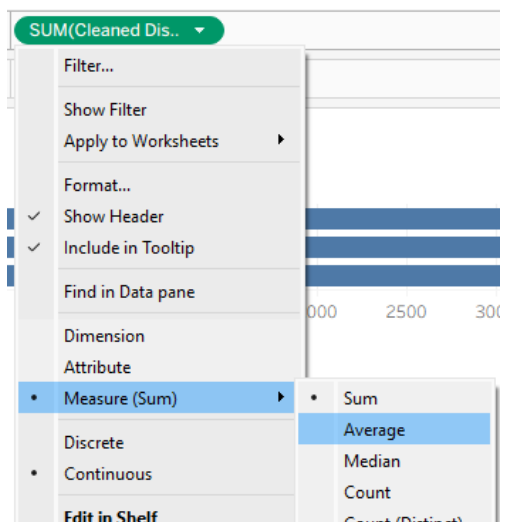
OK

Columns

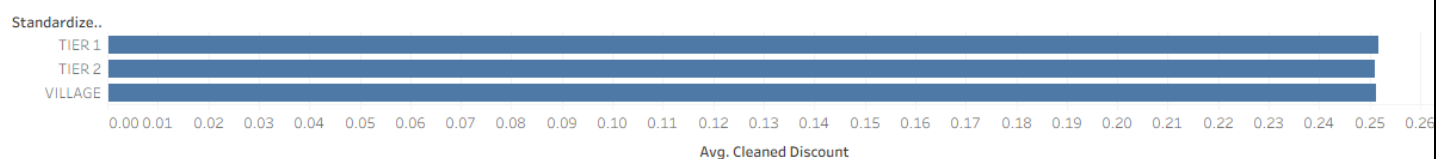
AVG(Cleaned Discou..

Rows

Standardized City Ty..



Standardized City Type & Cleaned Discount T054



CONCLUSION:

The practical was successfully completed using Tableau. We created calculated fields, performed row-level and aggregate-level calculations, and used these calculations to analyze the sales data. The practical helped in understanding how Tableau calculations can be used to derive meaningful insights from the dataset and improve data analysis.